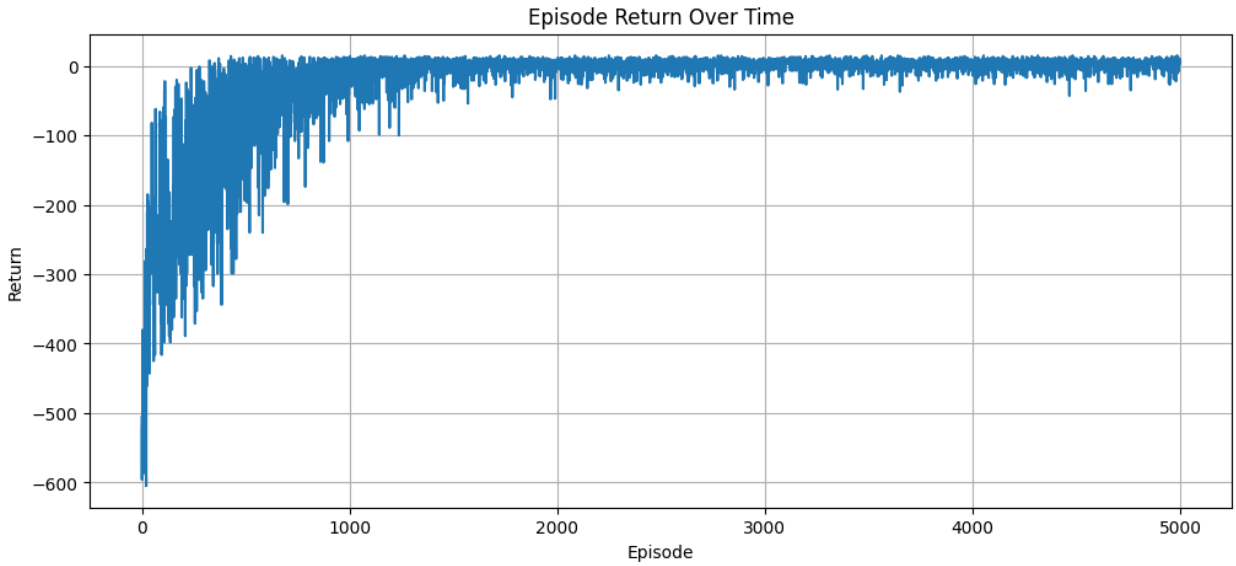
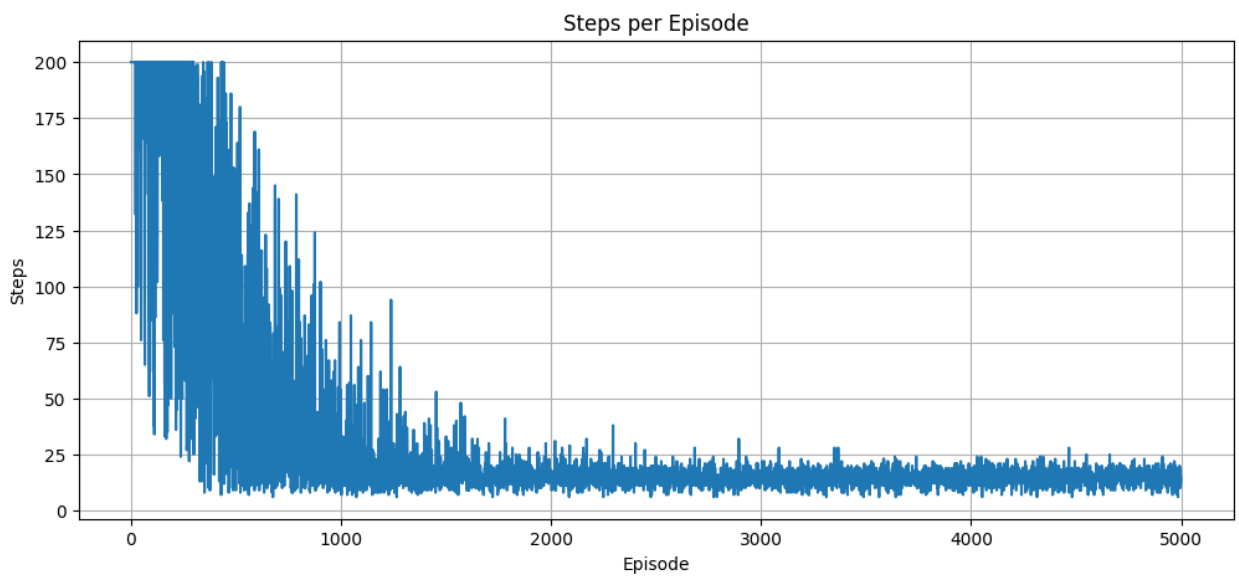
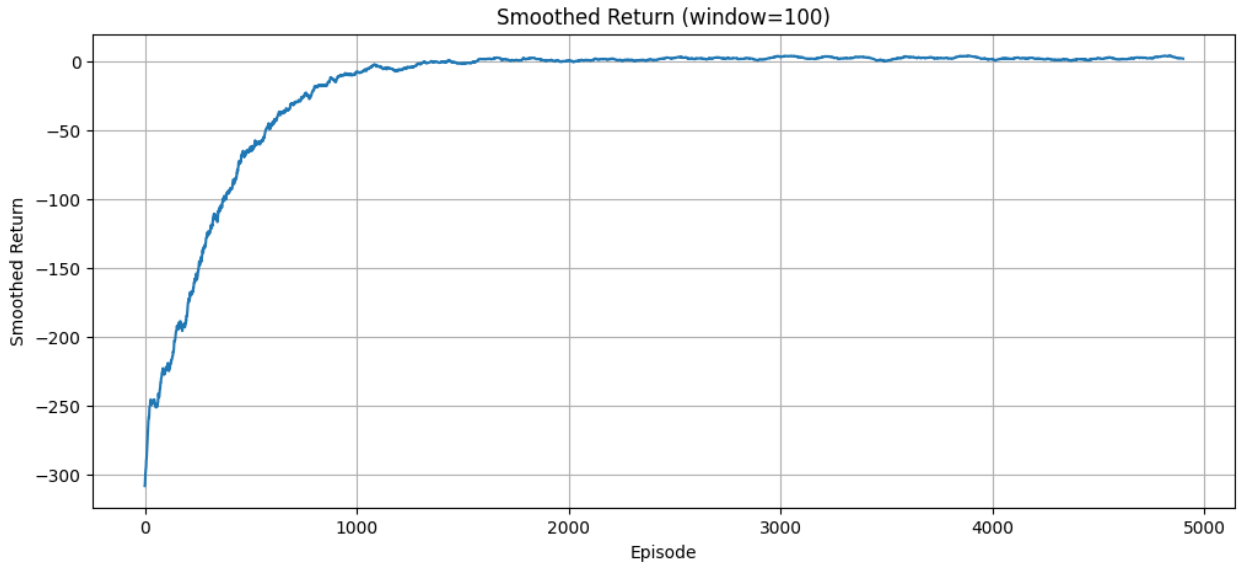
Assignment 2 Report

Plots/Graphs for initial training

- **Learning Rate α : 0.1**
- **Exploration Factor ϵ: 0.1**
- **Discount Factor γ : 0.9**

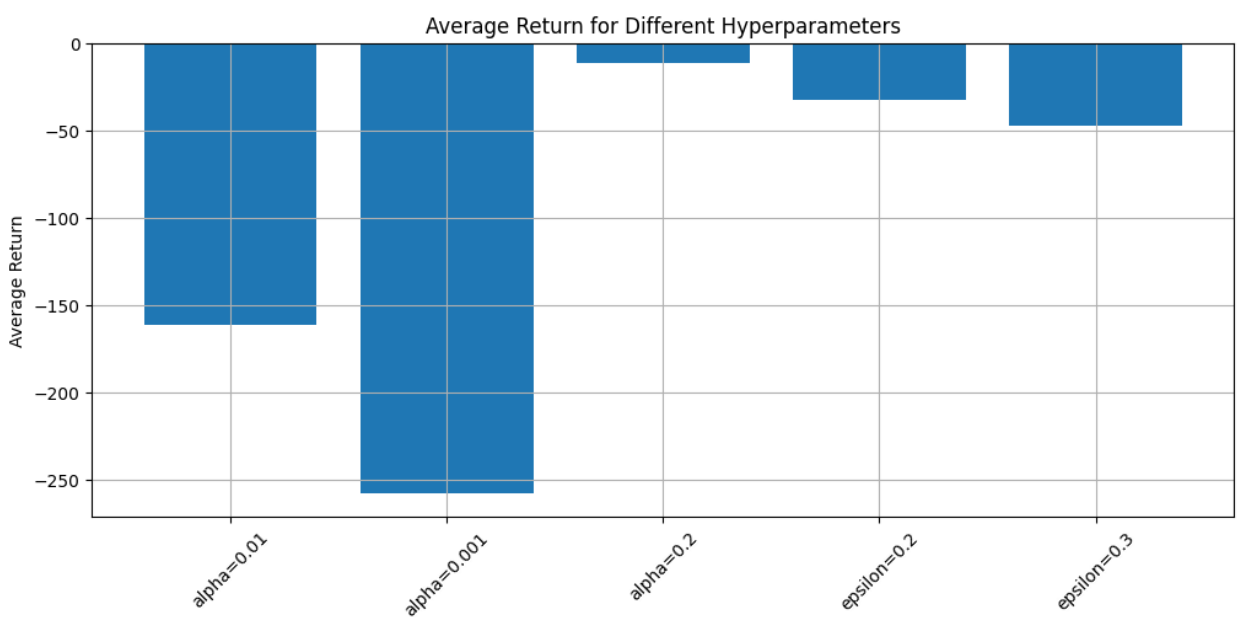


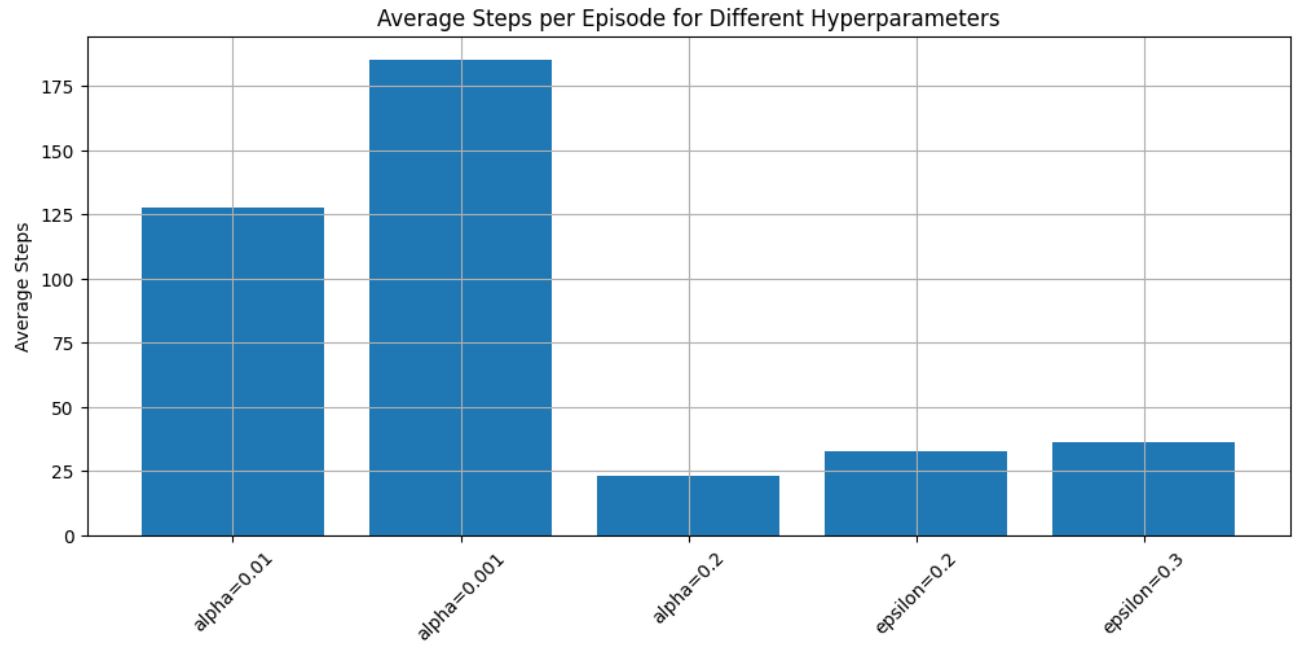


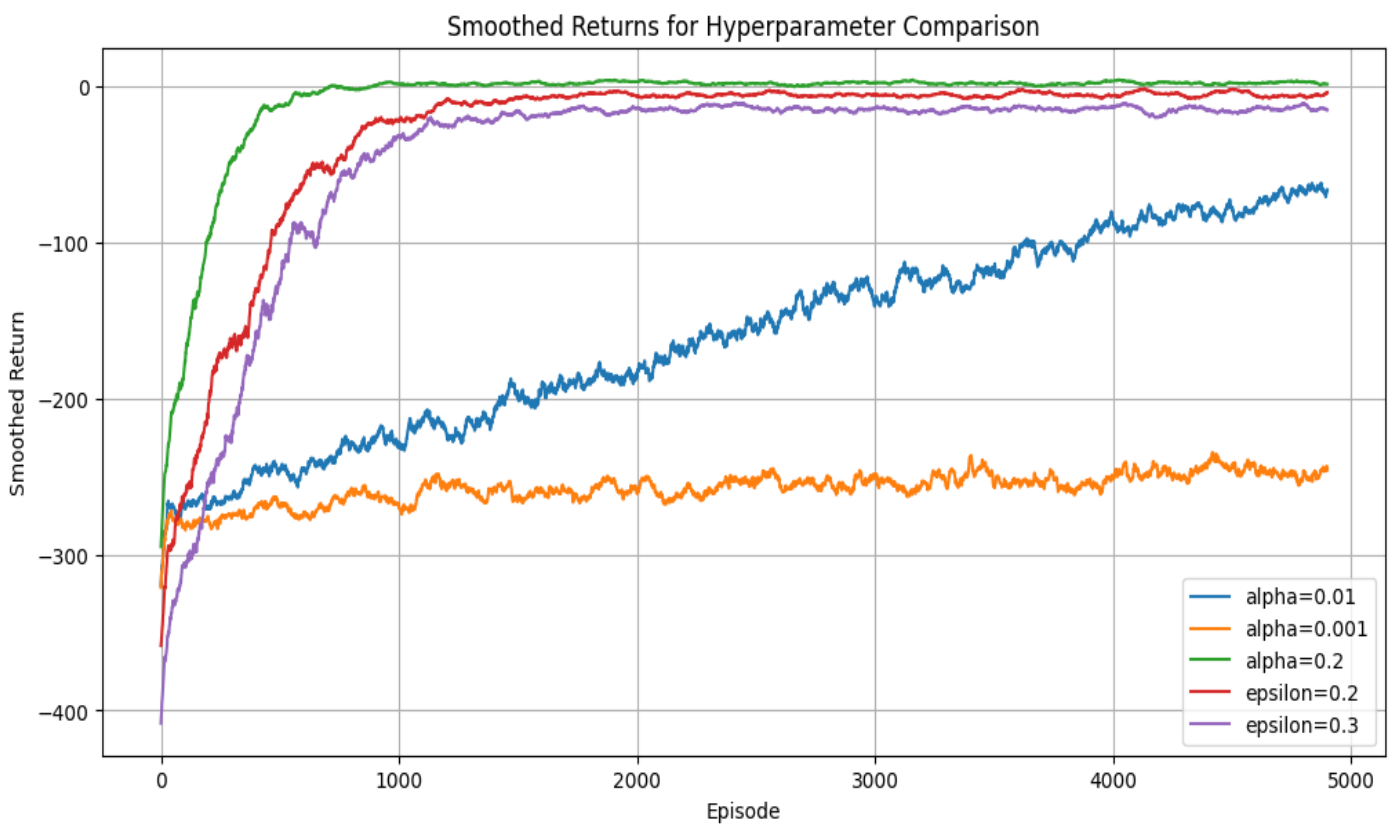


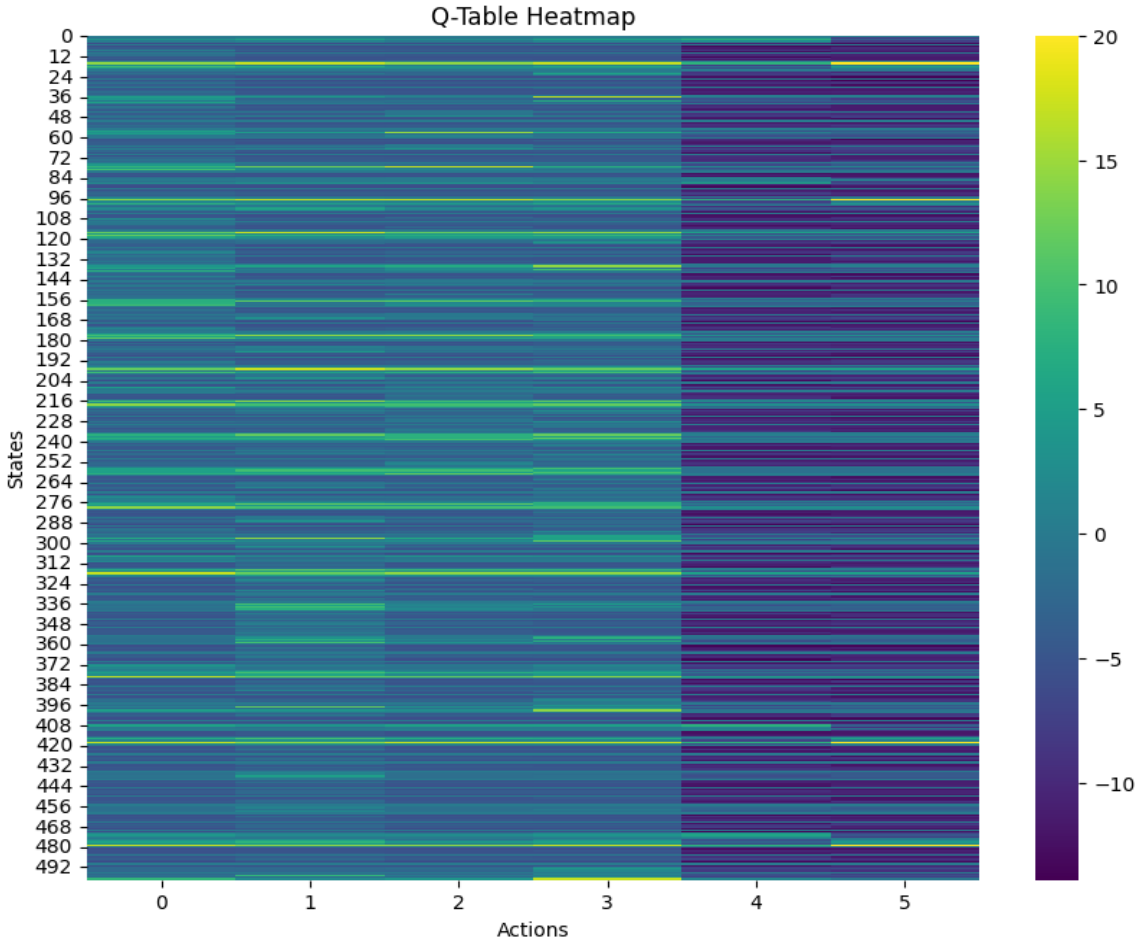
Plots/Graphs using different hyperparameters

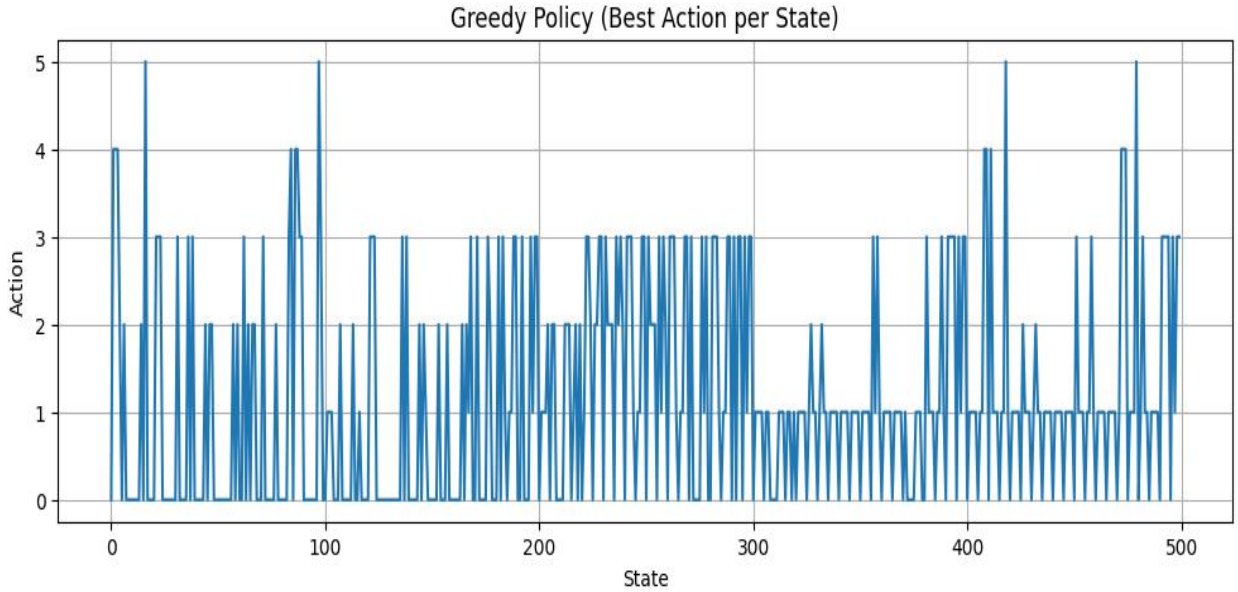
- Learning Rate α = [0.01, 0.001, 0.2]
- Exploration Factor γ = [0.2, 0.3]




Average Steps per Episode for Different Hyperparameters
Average Steps
0
25
50
75
100
125
150
175
alpha=0.01
alpha=0.001
alpha=0.2
epsilon=0.2
epsilon=0.3


Smoothed Returns for Hyperparameter Comparison
Smoothed Return
0
−100
−200
−300
−400
0
1000
2000
3000
4000
5000
Episode
alpha=0.01
alpha=0.001
alpha=0.2
epsilon=0.2
epsilon=0.3


Q-Table Heatmap
States
Actions
0
12
24
36
48
60
72
84
96
108
120
132
144
156
168
180
192
204
216
228
240
252
264
276
288
300
312
324
336
348
360
372
384
396
408
420
432
444
456
468
480
492
0
1
2
3
4
5
20
15
10
5
0
−5
−10


Greedy Policy (Best Action per State)
Action
State
0
1
2
3
4
5
0
100
200
300
400
500

My Observations

Based on the graphs, I think the best combination of learning rate and exploration factor are Learning rate = 0.2, exploration rate = 0.2. On the bar charts for average steps per episode, these 2 values are the lowest, which is what we want. On the other bar chart for average return, these 2 values are the highest. On the smoothed returns chart (smoothed curved lines), these 2 values performed the best.